

weeks-1-and-2\graphs-functions.py

```
1 '''
2 next problem
3 '''
4     f(x) = 4x2 + (-1)
5 '''
6 function translation: [vertical] →
7     moving upward → add units to the end of the function equation
8     moving downward → subtract units from end
9
10 function translation: [horizontal] →
11     moving left → add units inside
12     moving right → subtract units inside
13 '''
14     h(x) = 4x2 + (-1) + 3
15     [h(x) = 4x2 + 2]
16
17
18 '''
19 next problem
20 '''
21     vertical upward 4-units
22
23     h(x) = (-2x2) + (-9)
24     f(x) = (-2x2) + (-9) + 4
25     [f(x) = (-2x2) + (-5)]
26
27
28 '''
29 next problem
30 '''
31     vertical downward 4-units
32
33     f(x) = (-3x2) + 5
34     f(x) = (-3x2) + 5 + (-4)
35     [h(x) = (-3x2) + 1]
36
37
38 '''
39 next problem
40 '''
41     (x = (-1))
42     (x = 1)
43     y = (-2)x2
44     [0, 0]
45 #vertex is (0,0) because no numbers are added or subtracted after the x2
46
47     y = (-2)(-1)2
48     y = (-2)(1)2
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49     y = (-2)(1)
50     y = (-2)
51     [(-1), (-2)]
52
53     y = (-2)(1)2
54     y = (-2)(1)
55     y = (-2)
56     [(1), (-2)]
57
58
59     y = (-2)(2)2
60     y = (-2)(4)
61     y = (-8)
62     [(2), (-8)]
63
64     y = (-2)(-2)2
65     y = (-2)(-8)
66     [(-2), (-8)]
67
68     '''
69     next problem
70     '''
71     y = x2
72
73 #vertex is (0,0) because no numbers are added or subtracted after the x2
74 '''
75 right 1, up 1,3,5 shortcut
76 '''
77     [0, 0]
78     [1, 1]
79     [2, 4]
80     [3, 9]
81
82 mirror the points across the vertical center line
83 '''
84     [-1, 1]
85     [-2, 4]
86     [-3, 9]
87
88 '''
89 next problem
90 '''
91 '''
92 right 1, up (1/3),(3/3),(5/3) shortcut
93 '''
94     y = (1/3)x2
95     [0, 0]
96     [1, (1/3)]
97     [2, 4/3]
98     [3, 9/3]
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99     [-1, (1/3)]
100    [-2, 4/3]
101    [-3, 9/3]
102
103    '''
104    next problem
105    '''
106    #translate the graph of f(x) to y = f(x) + k (vertical translation) or y = f(x - h)
(horizontal translation)
107    '''
108    function translation: [vertical] →
109        moving upward → add units to the end of the function equation
110        moving downward → subtract units from end
111
112    function translation: [horizontal] →
113        moving left → add units inside
114        moving right → subtract units inside
115    '''
116
117    y = f(x) + 2
118    #it's addition of the constant, so → vertical translation upward by 2 units for each
point on the graph
119
120    y = g(x + 5)
121    #it's addition inside the argument, so → horizontal translation to the left by 5 units
for each point on the graph
122
123    '''
124    next problem
125    '''
126    y = f(x) + (-4)
127    #it's subtraction of the constant, so → vertical translation downward by 4 units for
each point on the graph
128
129    y = g(x + 2)
130    #it's addition inside the argument, so → horizontal translation to the left by 2 units
for each point on the graph
131
132    '''
133    next problem
134    '''
135    y = f(x - 5)
136    #it's subtraction inside the argument, so → horizontal translation to the right by 5
units for each point on the graph
137
138    y = g(x) + (-2)
139    #it's subtraction of the constant, so → vertical translation downward by 2 units for
each point on the graph
140    '''
141    next problem
142    '''

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143    y = |x + (-3)| + 1
144    #it's subtraction inside the argument, so → horizontal translation to the right by 3
units for each point on the graph
145    #it's addition of the constant, so → vertical translation upward by 1 unit for each
point on the graph
146    '''
147    next problem
148    '''
149    y = (x + (-4))3 + (-3)
150    #it's subtraction inside the argument, so → horizontal translation to the right by 4
units for each point on the graph
151    #it's subtraction of the constant, so → vertical translation downward by 3 units for
each point on the graph
152    #the cube only affects the shape of the graph, not the translation
153    '''
154    next problem
155    '''
156    y = -(x + 4) + (-1)
157    #it's addition inside the argument, so → horizontal translation to the left by 4 units
for each point on the graph
158    #it's subtraction of the constant, so → vertical translation downward by 1 unit for each
point on the graph
159    '''
160    next problem
161    '''
162    f(x) = (x + 3) / (3x + (-5))
163    find f(a + (-2))
164
165    = ((a + (-2)) + 3) / (3(a + (-2)) + (-5))
166    = (a + 1) / (3a + (-6) + (-5))
167    = (a + 1) / (3a + (-11))
168    '''
169    next problem
170    '''
171    h(x) = x + 1
172    find (h)(y + 5)
173
174    = (y + 5) + 1
175    = y + 6
176    '''
177    next problem
178    '''
179    f(x) = 7 + 2x / 3 + x
180    find f(5x)
181
182    = 7 + 2(5x) / 3 + (5x)
183    = (7 + 10x) / (3 + 5x)
184
185    #Since 7, 10x, 3, and 5x do not share a common factor that can be factored out of the
entire top and bottom, the fraction is in its simplest form
186    '''

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187 next problem
188 '''
189     #a relation is a function if each input (domain) maps to exactly one output (range)
190 '''
191 next problem
192 '''
193     #if any vertical line passes through more than one point at the same time on the graph,
194     it's not a function
195 '''
196 next problem
197 '''
198     f(x) = y
199 '''
200 next problem
201 '''
202     #endpoints with solid dots → use []
203     #endpoints with empty dots → use ()
204
205     'domain' = [](-2), 3()
206     'range' = [](-3), 5()
207 '''
208 next problem
209 '''
210     f(x) = (4 + (x2)) / (4 + x)
211     find f(4)
212
213     f(4) = (4 + (42)) / (4 + 4)
214     f(4) = (4 + 16) / 8
215     f(4) = 20 / 8
216         / 4 / 4
217 #-----
218     [f(4) = 5 / 2]
219
220
221     g(x) = 3 + √(x + 12)
222     find g(-3)
223
224     g(-3) = 3 + √((-3) + 12)
225     g(-3) = 3 + √(9)
226     g(-3) = 3 + 3
227     [g(-3) = 6]
228
229     h(x) = | ((-1) / 4)x + (-16) |
230     find h(16)
231
232     h(16) = | ((-1) / 4)(16) + (-16) |
233     h(16) = | (-4) + (-16) |
234     h(16) = | -20 |
235     [h(16) = 20]

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236
237     #multiplying a whole number by a fraction → the same as dividing the whole number by the
238     denominator of the fraction and then multiplying it by the numerator
239
240 next problem
241 '''
242     f(x) = (3x2 + (-10)) / x2
243     find f(4)
244
245     = (3(4)2 + (-10)) / (4)2
246     = (3(16) + (-10)) / 16
247     = (48 + (-10)) / 16
248     = 38 / 16
249         / 2 / 2
250 #-----
251     [f(4) = 19 / 8]
252
253     g(x) = 4|x + (-11)|
254     find g(2)
255
256     = 4|2 + (-11)|
257     = 4|-9|
258     = 4(9)
259     [g(2) = 36]
260
261     h(x) = (-6) + √(x + (-1))
262     find h(4)
263
264     = (-6) + √(4 + (-1))
265     = (-6) + √(3)
266     [h(4) = (-6) + √(3)]
267
268 '''
269 next problem
270 '''
271     f(x) = √(4 + (-4x))
272     find f(-3)
273
274     = √(4 + (-4(-3)))
275     = √(4 + 12)
276     = √(16)
277     [f(-3) = 4]
278
279     g(2) = (x3 + 9) / x3
280     find g(2)
281
282     = (23 + 9) / 23
283     = (8 + 9) / 8
284     = 17 / 8

```

```
285     [g(2) = 17 / 8]
286
287     h(x) = |(-2 / 3)x + (-13)|
288     find h(12)
289
290     #multiplying a whole number by a fraction → the same as dividing the whole number by the
denominator of the fraction and then multiplying it by the numerator
291     = |(-2 / 3)(12) + (-13)|
292     #12 ÷ 3 = 4
293     #4 * (-2) = (-8)
294
295     = |(-8) + (-13)|
296     = |-21|
297     [h(12) = 21]
298
299
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301
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